

VORONEZH CHERTOVITSKY, Russian Federation

WMO: 341220

Lat: 51.814N

Lon: 39.230E

Elev: 514

StdP: 14.42

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-5.7	-17.0	2.2	-11.4	-11.0	3.1	-5.7	26.8	29.4	23.8	26.6	5.1	310	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	91.4	66.4	87.6	65.5	84.0	64.5	69.5	82.1	68.0	80.8	66.8	79.1	8.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.0	73.7	64.2	91.8	72.1	62.6	86.6	71.0	33.8	81.9	32.7	81.0	31.6	78.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.5	18.6	16.3	DB	-18.1	96.1	7.3	3.7	-23.4	98.7	-27.6	100.9	-31.7	103.0	-37.1	105.7
(5)				WB	-17.2	73.0	7.3	2.6	-22.5	74.9	-26.7	76.4	-30.8	77.8	-36.1	79.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	19.2	20.0	30.0	46.5	59.6	64.8	69.3	68.3	56.9	44.9	33.9	26.3	
	DBStd	19.99	12.68	12.23	9.47	8.16	7.94	6.99	6.04	7.16	6.78	8.26	8.22	9.55	
	HDD50	4030	956	839	620	162	14	1	0	0	17	201	486	734	
	HDD65	7693	1421	1259	1085	555	203	90	24	45	255	623	934	1199	
	CDD50	2247	0	0	1	57	312	444	599	566	223	43	2	0	
	CDD65	434	0	0	0	0	36	83	158	146	11	0	0	0	
	CDH74	4926	0	0	0	17	522	894	1607	1705	181	0	0	0	
	CDH80	1767	0	0	0	1	133	274	603	722	34	0	0	0	
Wind	WSAvg	8.2	8.8	9.4	9.2	8.6	7.6	7.6	6.8	6.8	7.1	8.0	8.7	9.5	
Precipitation	PrecAvg	22.3	1.5	1.4	1.3	1.6	1.9	2.5	2.3	2.2	2.2	2.0	1.7	1.7	
	PrecMax	31.5	3.3	2.6	2.5	5.8	4.2	4.7	4.5	7.2	6.9	5.0	4.3	4.4	
	PrecMin	15.9	0.6	0.2	0.0	0.2	0.2	0.7	0.2	0.2	0.2	0.4	0.1	0.6	
	PrecStd	4.1	0.6	0.6	0.7	1.2	1.0	1.1	1.1	1.4	1.7	1.1	1.2	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	41.1	58.9	76.7	88.0	91.5	96.4	99.0	84.3	70.0	56.9	46.6	
		MCWB	39.8	39.2	46.3	55.2	65.3	65.6	65.9	65.7	60.3	55.8	52.1	45.5	
	2%	DB	37.5	37.7	51.6	70.2	83.9	86.4	90.0	92.9	79.1	64.8	52.1	43.0	
		MCWB	36.5	36.1	42.8	52.9	62.4	64.7	66.8	66.0	59.3	55.1	48.8	42.3	
	5%	DB	35.5	35.8	46.3	66.1	80.3	82.6	86.3	87.9	74.9	61.0	48.1	40.7	
		MCWB	34.6	34.4	40.8	51.7	61.0	65.2	66.4	65.6	58.9	54.0	45.5	39.8	
	10%	DB	33.7	34.1	42.4	61.1	75.6	79.0	82.5	82.7	69.9	57.1	44.6	37.5	
		MCWB	33.0	33.1	38.3	48.6	59.4	63.7	65.0	64.1	57.3	51.3	43.2	36.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	39.6	47.9	57.6	67.7	71.0	71.3	70.9	65.0	60.0	53.2	45.5
			MCDB	40.1	40.1	56.9	71.5	83.7	80.7	85.1	84.0	72.6	65.4	56.2	46.3
2%		WB	36.4	36.6	43.7	54.9	64.5	68.7	69.6	68.5	62.0	57.1	49.5	42.0	
		MCDB	36.9	37.3	49.0	65.7	77.9	79.4	82.5	83.2	72.0	62.4	52.1	42.5	
5%		WB	34.2	34.4	40.9	52.9	62.6	67.1	68.2	67.1	60.1	54.6	45.5	39.3	
		MCDB	34.6	35.0	44.8	63.1	75.8	77.8	80.7	81.9	70.5	60.3	47.2	39.9	
10%		WB	32.6	33.2	38.3	50.5	60.8	65.3	66.9	65.6	58.4	52.4	42.9	36.3	
		MCDB	33.1	33.9	41.5	59.1	71.8	75.6	78.4	79.5	68.6	57.0	43.9	37.0	
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.7	14.0	19.5	22.1	20.7	21.1	23.1	20.0	14.4	9.3	8.2	
		MCDBR	6.8	7.9	19.2	25.9	28.1	25.8	26.2	28.9	27.6	20.6	14.2	7.8	
		MCWBR	6.5	7.2	12.1	13.2	11.5	10.4	9.1	9.7	12.3	12.3	11.5	7.6	
	5% WB	MCDBR	6.6	7.5	16.8	22.9	23.9	21.7	21.5	24.0	22.4	17.7	11.2	7.4	
		MCWBR	6.6	7.3	11.1	12.8	11.6	10.4	9.0	10.4	11.7	11.8	10.0	7.3	
Clear-Sky Solar Irradiance	taub	0.304	0.340	0.364	0.440	0.396	0.389	0.420	0.436	0.388	0.361	0.338	0.306		
	taud	2.124	2.028	2.098	2.090	2.317	2.354	2.276	2.216	2.334	2.353	2.276	2.175		
	Ebn at Noon	212	236	258	251	272	275	263	251	251	233	201	190		
	Edh at Noon	24	35	40	46	38	37	40	40	32	26	21	20		
All-Sky Solar Radiation	RadAvg	300	627	1065	1325	1814	1886	1849	1599	1057	555	243	172		
	RadStd	69	83	170	117	147	168	153	137	140	78	50	49		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	-347	+223	N/A

Nomenclature: See separate page